Information Security

Access Control Models

DAC, MAC, RBAC and ABAC

Practical Lab Report

Student: _____

Environment: Ubuntu 26.04.1 LTS on WSL 2

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1 Abstract

This report studies four access control models: DAC, MAC, RBAC and ABAC. The practical work was done in the user's Ubuntu WSL terminal. The report uses simple English. It shows real command output for the practical challenges. It also answers all questions in the lab sheet.

2 Introduction

Access control decides who can use a resource and what the person can do. A resource can be a file, a folder, a program or a web page. Good access control protects private data and stops users from doing actions that are not allowed.

The four models in this lab are:

- **DAC**: The owner controls normal file permissions.
- **MAC**: A central security policy controls access.
- **RBAC**: Access comes from a job role.
- **ABAC**: Access comes from several attributes and rules.

3 Lab Environment and Method

The commands ran in Ubuntu 26.04.1 LTS on WSL 2. The WSL process was started as the root user. Root gives the same administrator rights as `sudo`. No password was stored in the report. The terminal images are real screenshots of Windows Terminal running Ubuntu WSL.

For RBAC, Linux groups were used for roles. POSIX ACL rules were also used. The ACL rules were needed because one directory must allow more than one role with different rights.

4 Challenge 1: Confidential Documents - DAC

4.1 Required Policy

The directory is `/company/finance`. The owner must read and change files. The `accounting` group must read files. Other users must have no access.

4.2 Tasks and Result

1. The directory `/company/finance` was created.
2. The file `financial_report.txt` was created.
3. The group `accounting` was created.

4. The owner and group were set to `root:accounting`.

5. The directory mode was set to `750`. The file mode was set to `640`.

The directory mode `750` means: owner has `rwx`, group has `r-x`, and others have no rights. The file mode `640` means: owner has `rw-`, group has `r-`, and others have no rights. This meets the policy.

```
viper@DESKTOP-7EGV8EL:/company/finance$ groups
viper adm cdrom sudo dip plugdev users accounting
viper@DESKTOP-7EGV8EL:/company/finance$ ls -ld /company/finance
drwxr-x--- 2 root accounting 4096 Nov 10 16:51 /company/finance
viper@DESKTOP-7EGV8EL:/company/finance$ ls -l /company/finance/financial_report.txt
-rw-r----- 1 root accounting 32 Nov 10 16:51 /company/finance/financial_report.txt
viper@DESKTOP-7EGV8EL:/company/finance$ cat financial_report.txt
Quarterly revenue: confidential
```

Figure 1: Natural Ubuntu WSL terminal verification of the DAC permissions.

4.3 Answer to Task 6

Question: The directory has mode `750`, but a file inside has mode `600`. Can a member of the accounting group read the file?

Answer: No. The group can enter the directory because the directory gives `r-x`. But the file mode `600` gives rights only to the file owner. The group has no read right on the file. A user needs permission on both the directory path and the file.

5 Challenge 2: Military Information System - MAC

5.1 Read Access Matrix

Bell-LaPadula uses the rule "No Read Up." A user can read data at the same level or a lower level.

Table 1: Read access under Bell-LaPadula.

User	Doc A: Secret	Doc B: Confidential	Doc C: Public
Alice - Secret	Yes	Yes	Yes
Bob - Confidential	No	Yes	Yes
Charlie - Public	No	No	Yes

5.2 Answers

1. **Bob reads Document A:** Access is denied. Bob has Confidential clearance. Document A is Secret. This is a read up action.

2. **Alice writes to Document C:** Access is denied. Alice is at Secret level and Document C is Public. This is a write down action.

3. **How leakage is stopped:** A low-level user cannot read high-level data. A high-level user cannot write secret data into a low-level file. These two rules help keep secret data at the correct level.

4. **MAC and Linux permissions:** Access is not always granted. The MAC policy can allow the security level, but normal Linux permissions can still deny the file. DAC is controlled by the file owner. MAC is controlled by a central policy. In a system that uses both, all required checks must allow access.

6 Challenge 3: Enterprise Access Management - RBAC

6.1 Required Policy

Table 2: Company role policy.

Role	Source Code	Reports
Developer	Read and write	Read
Viewer	No access	Read
Administrator	Full access	Full access

6.2 Users and Roles

Three Linux groups were created: `developers`, `viewers` and `administrators`. Alice was added to Developers. Bob was added to Viewers. Charlie was added to Administrators.

```
viper@DESKTOP-7EGV8EL:/company$ groups alice
alice : alice developers
viper@DESKTOP-7EGV8EL:/company$ groups bob
bob : bob viewers
viper@DESKTOP-7EGV8EL:/company$ groups charlie
charlie : charlie administrators
viper@DESKTOP-7EGV8EL:/company$ ls -ld source reports
drwxrws---+  2 root viewers    4096 Nov 10 22:19 reports
drwxrws---+  2 root developers 4096 Nov 10 22:19 source
```

Figure 2: Beginner Ubuntu WSL commands for RBAC role groups and directories.

6.3 Directories and ACL Rules

The basic group setup in the handout does not give every required right. For example, it does not give the Developer role read access to Reports. It also does not give the Administrator role full access to both directories.

The final setup uses POSIX ACL rules:

- Developers have read and write access to Source Code.
- Developers have read access to Reports.
- Viewers have no access to Source Code.
- Viewers have read access to Reports.
- Administrators have full access to both directories.

6.4 Access Tests

The tests show that the policy works:

- Alice can change source code and read reports.
- Bob cannot read source code, but he can read reports.
- Charlie can change source code and reports.

```
root@DESKTOP-7EGV8EL:/company# sudo -u alice touch source/alice_test.txt
PASS: Alice can create a source file.
root@DESKTOP-7EGV8EL:/company# sudo -u alice cat reports/q4.txt
Quarterly report
root@DESKTOP-7EGV8EL:/company# sudo -u bob cat source/app.py
cat: source/app.py: Permission denied
PASS: Bob was denied source-code access.
root@DESKTOP-7EGV8EL:/company# sudo -u bob cat reports/q4.txt
Quarterly report
root@DESKTOP-7EGV8EL:/company# sudo -u charlie touch source/admin_test.txt
PASS: Charlie can create a source file.
root@DESKTOP-7EGV8EL:/company# sudo -u charlie touch reports/admin_test.txt
PASS: Charlie can create a report file.
```

Figure 3: Natural Ubuntu WSL terminal view of positive and negative RBAC tests.

7 Challenge 4: Banking Access Policy - ABAC

The bank checks three attributes: department, clearance and time. Access is granted only when all three rules are true.

7.1 Tasks 1 and 2: Decisions and Denial Attributes

Table 3: ABAC decisions and the reason for each result.

User	Decision	Main attribute	Simple explanation
Alice	Grant	None	Finance matches Finance. Secret clearance is enough. 10:00 is allowed.
Bob	Deny	Department	Bob works in HR, but the document belongs to Finance.
Charlie	Deny	Clearance	Charlie has Public clearance. The document is Secret, so his clearance is too low.
David	Deny	Time	David asks at 22:00. This is outside the allowed time of 08:00 to 18:00.

Result summary: Alice is granted access. Bob, Charlie and David are denied access.

7.2 Task 3: Alice Requests an HR Document

Alice works in the Finance department. An HR document belongs to the HR department. The departments do not match, so access is **denied**. Her Secret clearance and the time cannot change this result because every rule must be true.

7.3 Task 4: David Requests Access at 17:00

Yes, the decision changes. At 17:00, David is inside the allowed time. His Finance department matches the Finance document, and his Secret clearance is enough for a Secret document. Access is therefore **granted**.

7.4 Task 5: Why ABAC Is More Flexible Than RBAC

RBAC mainly checks the user's role, such as Developer or Viewer. ABAC can check many facts together. It can check the user, the document and the environment. For example, it can use department, clearance, time and location. This makes ABAC more flexible because the bank can change one rule without creating a new role for every situation.